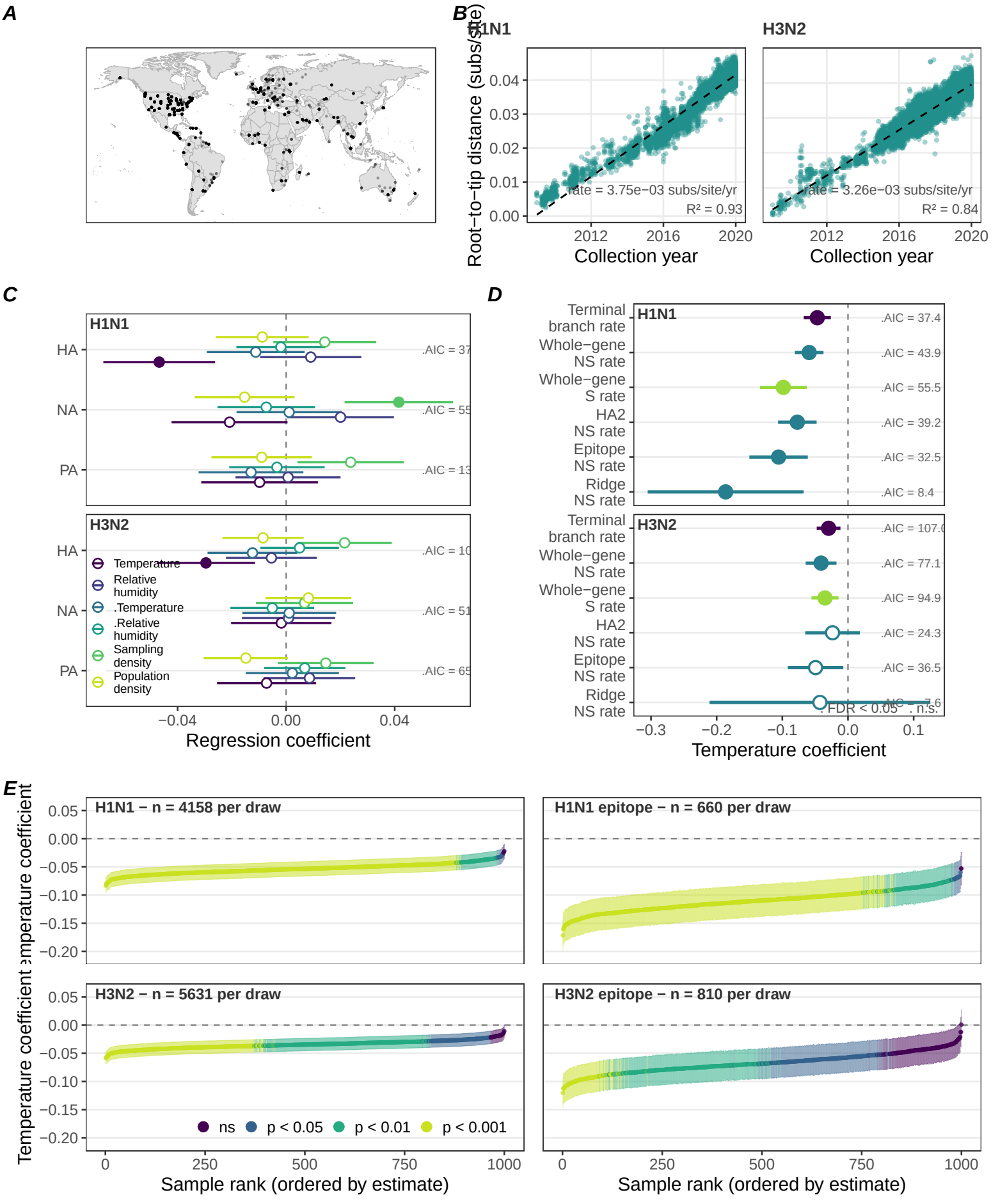

A
B
H1N1
H3N2
Root−to−tip distance (subs/site)
0.04
0.03
0.02
0.01
0.00
rate = 3.75e−03 subs/site/yr
R² = 0.93
rate = 3.26e−03 subs/site/yr
R² = 0.84
2012
2016
2020
Collection year
C
D
HA
NA
PA
.AIC = 37
.AIC = 55
.AIC = 13
.AIC = 10
.AIC = 51
.AIC = 65
Temperature
Relative humidity
.Temperature
.Relative humidity
Sampling density
Population density
−0.04
0.00
0.04
Regression coefficient
Terminal branch rate
Whole−gene NS rate
Whole−gene S rate
HA2 NS rate
Epitope NS rate
Ridge NS rate
.AIC = 37.4
.AIC = 43.9
.AIC = 55.5
.AIC = 39.2
.AIC = 32.5
.AIC = 8.4
.AIC = 107.0
.AIC = 77.1
.AIC = 94.9
.AIC = 24.3
.AIC = 36.5
.AIC = 7.6
. FDR < 0.05 . n.s.
−0.3
−0.2
−0.1
0.0
0.1
Temperature coefficient
E
H1N1 − n = 4158 per draw
H1N1 epitope − n = 660 per draw
H3N2 − n = 5631 per draw
H3N2 epitope − n = 810 per draw
Temperature coefficient
0.05
0.00
−0.05
−0.10
−0.15
−0.20
ns
p < 0.05
p < 0.01
p < 0.001
0
250
500
750
1000
Sample rank (ordered by estimate)